

Testing Arguments by Machine: An Introduction to Truth Tables

An original lesson plan inspired by **John Sanders's** the free video course *Symbolic Logic: A Free Online Course* (Open Culture). Not affiliated with or endorsed by Open Culture or John Sanders. Explore the original course: <https://www.openculture.com/symbolic-logic-a-free-online-course>

GRADE LEVEL

Grades 11-12 / Intro College

DURATION

65 minutes

TOPICS

Logic

“How can we mechanically test whether an argument is valid, using nothing but a grid of true/false combinations, without ever relying on our gut sense of what 'sounds right'?”

Learning Objectives

By the end of this lesson, students will be able to:

- Translate simple English sentences into symbolic form using the connectives for negation, conjunction, disjunction, and the conditional.
- Construct a complete truth table for a compound statement and correctly compute its truth value under every possible assignment of truth values to its parts.
- Use a truth table to test whether a two-premise argument is valid, by checking whether any row makes the premises true and the conclusion false.

Materials Needed

- Handout listing the five basic connectives (not, and, or, if-then, if-and-only-if) with their symbols and truth-value definitions
- Blank truth table templates sized for two- and three-variable statements
- Whiteboard for building one full truth table together as a class
- Set of argument cards in English for students to symbolize and test using truth tables

Lesson Procedure

Hook: Is 'Or' Always Inclusive?

8 min

Ask students what 'you can have cake or ice cream' means versus 'you need a passport or a driver's license to board.' Get them to notice ordinary English 'or' is ambiguous between at-least-one (inclusive) and exactly-one (exclusive), and that logic needs to pick one meaning and stick to it. Introduce that symbolic logic solves this ambiguity by giving each connective one fixed, unambiguous meaning defined entirely by a table of truth values.

Direct Instruction: The Five Connectives

15 min

Introduce negation, conjunction, disjunction (defined as inclusive), the conditional, and the biconditional, giving the truth-value definition of each. Pause on the conditional, since students find it most counterintuitive (it is false only when the antecedent is true and the consequent is false), and give a concrete everyday example to anchor it.

Guided Practice: Build a Table Together

15 min

As a class, symbolize a compound sentence and build its full truth table row by row on the board, walking through how compound truth values are computed from their parts, column by column, left to right.

Small-Group Work: Test an Argument

20 min

In pairs, students symbolize a short two-premise valid argument and one with a subtle flaw such as affirming the consequent, then build the truth table for each and determine validity by checking every row for a case where the premises are all true but the conclusion is false.

Closure: What Did the Table Prove?

7 min

Bring the class back together and ask, for the invalid argument, which specific row exposed the flaw. Emphasize that a truth table doesn't just give an answer, it gives a specific counterexample row that shows exactly how the premises could be true while the conclusion is false.

Discussion Questions

- 1 Why does symbolic logic define 'or' as strictly inclusive, and what would be lost or gained if it instead used the exclusive sense from everyday speech?
- 2 The conditional is only false in one specific case. Does this match how you naturally use 'if...then' in conversation, or does it feel like it's missing something?
- 3 What does it mean, mechanically, for a truth table to prove an argument invalid? What exactly are you pointing to in the table?
- 4 As the number of variables in a statement grows, the truth table grows exponentially. What does this suggest about the limits of using truth tables to test very complex arguments?
- 5 Affirming the consequent is a common and tempting mistake. Why might it feel intuitively convincing even though a truth table shows it's invalid?

Assessment

Students receive two new two-premise arguments in English, symbolize each into logical notation, build a complete truth table for each, and state a validity verdict, explicitly identifying the row that either confirms validity or demonstrates invalidity by naming the specific truth-value assignment that breaks it. Grade for correct symbolization, accurate table construction, and correct identification of the decisive row.

Extension Activity



Have students take a three-variable argument involving 'and,' 'or,' and 'if-then' together from a real debate or logic puzzle they find online, build the larger eight-row truth table required, and write a short paragraph explaining why testing validity purely mechanically, row by row, is both a strength (it removes guesswork) and a limitation (it becomes unwieldy fast) of the truth-table method.

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